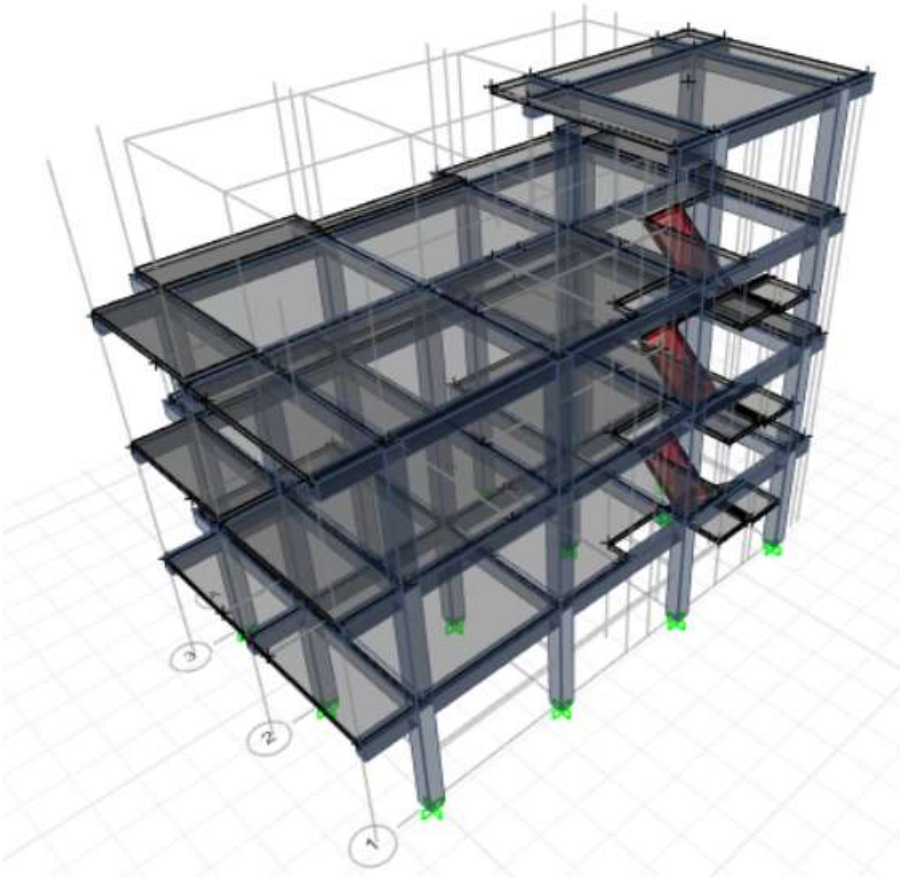
STRUCTURAL ANALYSIS

&

DESIGN REPORT

OF

RESIDENTIALBUILDING



……. CONSULTANCY AND CONSTRUCTION Pvt. Ltd.
Tulsipur Sub-Metropolitancity, Dang

Owner:-
Mrs. Radhika K.C. Khadka
Tulsipur-, Dang

2026 March

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ANNEX 1: LIST OF RELEVANT CODES

1. INTRODUCTION

1.1. BACKGROUND

The philosophy of seismic design can be summarized as follows:

1. Resist minor earthquakes without damage.
2. Resist moderate earthquake with minor structural and some non-structural damage.
3. Resist major catastrophic earthquakes without collapse.

Structures should be designed on **Weak Beam-Strong column** philosophy.

1.2. ABOUT THE REPORT

This report presents the structural engineering analysis and design of a Residential Building. This report describes in brief the structural aspects of the building, assumptions made, design process adopted, design standards complied with the proposed block. The analysis and design of the building have been carried out using a finite element software ETABS Ultimate21.0.0.

This report summarizes the structural analysis and design of residential located in Dang. It has planned to utilize the building as a Residential aspect**.** The analysis and design have been based on the prevailing codes that are in practice in Nepal, the National Building code of Nepal (105:2020) and the IS code at places if required. This report consists of the design procedures adopted, the assumptions made, the inputs made in the design and the design output.

1.3. PROPOSED STRUCTURAL SYSTEM

The structural system selected for the building is **RC buildings with Special Moment Resisting Frame (SMRF)**. All the partition and cladding walls are proposed to be constructed of brick masonry in cement mortar. The choice of structural system is influenced by the need of large open space for health facility rooms, the material and method of construction, the external architectural treatment, location and routing of service systems, the nature and magnitude of loading, and the height and height to width proportion of the building. Columns and beams have been laid out in the plan of the building in coordination with architectural and services planning. The structural system supports and transmits all forces arising from gravity, live and earthquake load from its point of origin to the ground. Attempts were made to provide the structure enough redundancy and an adequate reserve of strength and stiffness against premature failure, instability and degradation so the building performs as intended during the service life of the building.

1.4. STRUCTURAL ANALYSIS AND DESIGN PROCEDURE

The following steps were followed for analysis and design of the structure:

- Structural analysis and design of this building has been completed using ETABS, Software produced by CSI, California Berkeley;
- A 3-Dimensional finite element model was prepared using the software as per actual dimensions shown in the drawings. Center line dimensions are used for numerical model construction;

- Beams and columns are modeled as linear frame element whereas the slabs are modeled as “thin shell” elements;
- Floor slab is assumed to provide rigid floor diaphragm effect. The numerical model was constructed accordingly;
- Initially, sizes of structural components were proportioned based on the past experiences, which were later revised after the preliminary design;
- The dead loads were estimated by the software based on the member sizes. The finishes to the floors were applied to the floors. Wall proposed along beam were applied as line loads, however,load of wall over slab were assumed uniformly distributed on the floor;
- The imposed loads were applied to the floor and roof slabs;
- Seismic loads were considered acting in the horizontal direction (along either of the two principal directions) only. Seismic actions in vertical were ignored considering its insignificant effect on overall design;
- An earthquake is unlikely to occur simultaneously with maximum flood, wind, hence concurrency of flood, wind and earthquake loads were not considered in analysis of structural frame;
- The design seismic force has been estimated and automatically distributed by the software at various floor levels. The building elements have been designed for the critical actions derived using Equivalent Static Method (ESM);
- Under seismic force, the building has been analyzed in two principal directions.The principal lateral load resisting elements were designed and detailed as per NBC 105: 2020 standards; and
- The proposed structure was also checked against soft-story effect. To avoid any deficient effects such as short column due to the infill walls, the structure was also checked against these effects.

2. DETAILED PARAMETERS OF THE BUILDING

2.1. GENERAL INFORMATIONOF THE BUILDING

Owner's Name: Residential Building (**Mrs. Radhika K.C. Khadka)**
Location: Tulsipur, Dang
Plinth Area: 90.00 Sqm

2.2. PLAN OF THE BUILDING

Ground floor plan of the building is presented in Figure 1.

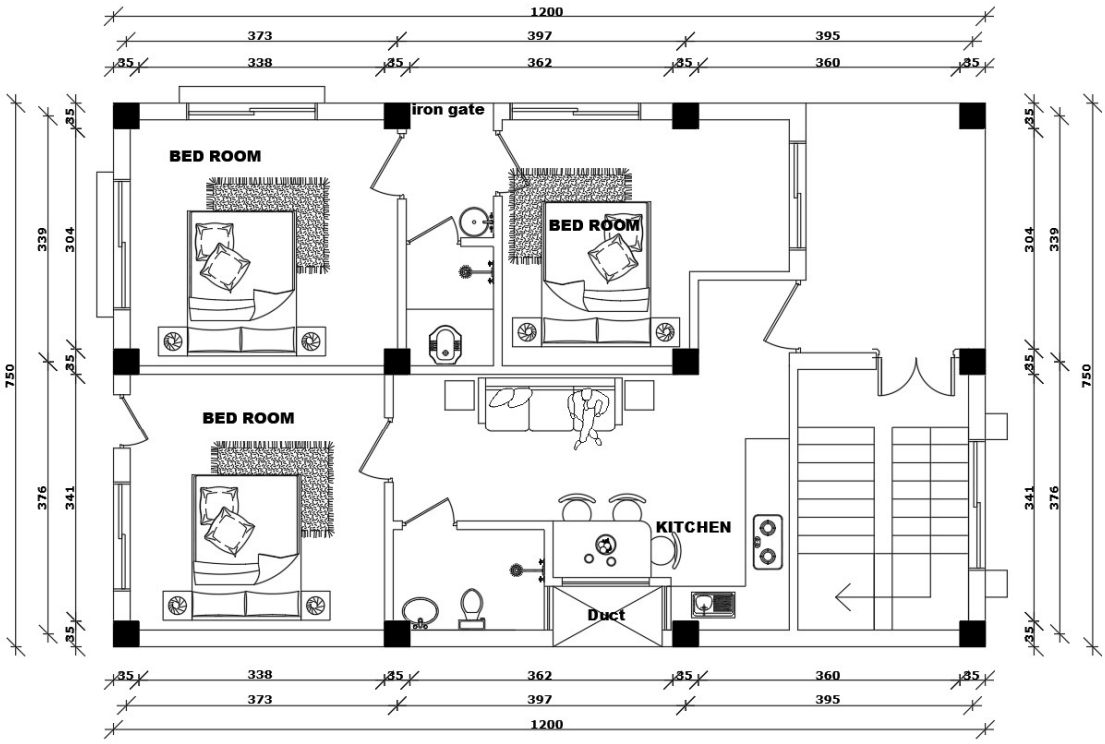


Figure 1: Plan of the building

2.3. General Parameters of The Building

2.3.1. Architectural and Functional Requirements

Building Occupancy:	Residential Building
No. of Story:	4storey
Design provision for future extension:	No
External brick walls thickness:	230mm
Internal brick wall thickness:	115 mm

2.3.2. Structural Parameters of The Building

Types of structural system: Reinforced Concrete Moment Resistant Frame.

Story Height:	3.125 m
Total height of Building:	12.625m
Slab Thickness:	125 mm
Footing Type:	Eccentric and isolated
Beam Sizes:	250*400 and 230*230
Secondary beam:	230*230 if required
Column Size:	350*350

Safe Bearing Capacity (SBC): 150 KN/m^2 (Assumed for Foundation Design)

Soil Type: Type D(for Dang Valley)

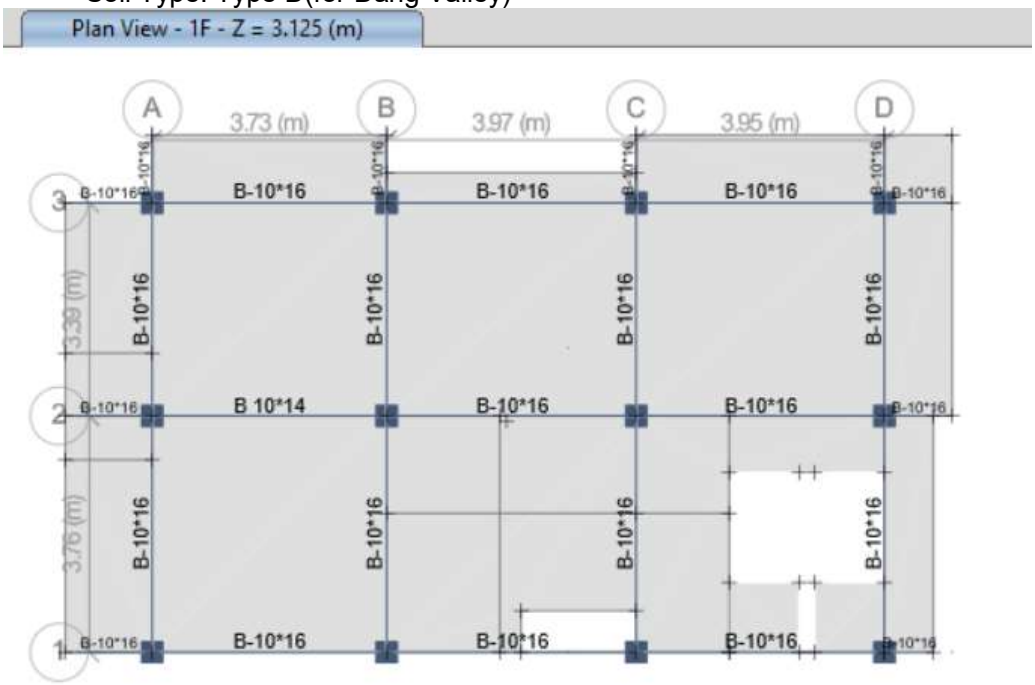


Figure 2 : Floor Beam Plan Layout

Figure 3 : Elevation of the building

2.3.3. MATERIAL PROPERTIES

Grade of Concrete: M20 for all structure members.
Grade of Steel: Fe 500
Compressive strength of brick: 7.5 MPa
Compressive strength of mortar: 7.5 MPa.
Poisson's ratio: 0.2

2.4. ANALYSIS AND DESIGN BASIS

Computer Aided Design: ETABS Ultimate21.0.0.
Analysis method for Earthquake resistant design: Equivalent Static Method
Level of design: Professionally Engineered Structure
Design Guideline: IS 456:2000; NBC 105:2020;NBC 205

The structural elements have been designed for critical load combination considering both states (Ultimate and Serviceable State). Type of design load cases considered:

Dead Load
Live Load
Seismic load

2.4.1. DESIGN METHODOLOGY

The building structure was analyzed as moment resisting space frame.
Indian Standard IS: 456-2000: Code of Practice for Plain and Reinforced Concretehas been followed for design of structural elements.

Limit State Methodfor safety and serviceability requirements associated with the design loads and design strengths of the materials has been used. These design loads and design strengths are obtained by applying partial safety factors for characteristic loads and strengths of the materials concrete and steel.

Limit States

The Limit State method of design covers the various forms of failure. There are several limit states at which the structure ceases to function, the most important among them are:

a) **Limit state of collapse or total failure of structure.**
It corresponds to the maximum load carrying capacity. This limit state corresponds to flexure, compression, shear and torsion.

b) **Limit state of serviceability**
It includes excessive deflection and excessive local damage. Excessive deflection adversely affects comfort and finishes. The excessive local damage results in cracking of concrete, which impairs the efficiency, or appearance of the structure.

3. LOADING

3.1. DEAD LOAD & LIVE LOAD

WALL LOAD CALCULATION

full brick wall thickness= 0.23m
half brick wall thickness= 0.115m
floor to floor height= 3.125m
beam depth= 0.355m
slab thickness= 0.15m
height of wall resting on beam= 3.125-0.355
= 2.770m
height of wall resting on slab= 3.125-0.15
= 2.975m
unit weight of brick masonry= 19.2KN/m³ **IS: 875 (Part 1)-1987,Table-1 Item no. 36**
parapet wall height= 0.75m

Wall	WL (without opening) Unit: KN/m		25% opening		Parapet wall load Unit:KN/m
	9"	4.5"	9"	4.5"	4.5"
resting on beam	=0.23*2.77*19.2 **=12.23**	=0.115*2.77*19.2 **=6.12**	=0.75*12.23 **=9.173**	=0.75*6.12 **=4.59**	=0.75*0.115*19.2 **1.656**
resting on slab	=0.23*2.975*19.2 **=13.14**	=0.115*2.975*19.2 **=6.57**	=0.75*13.14 **=9.855**	=0.75*6.57 **=4.928**	

AREA LOAD CALCULATION

(i) Floor Finish

thickness of screeding= 50mm
unit weight of screeding/plaster= 20 KN/m³ **IS: 875 (Part 1)-1987,Table-1 Item no. 42**
unit weight of flooring= 0.22 KN/m³ **IS: 875 (Part 1)-1987,Table-2 Item no. 7**
thickness of plaster on ceiling= 12.50mm
flooring= 10.00mm
area considered= 1 m²

Floor Finish load

(screeding)= 20*(50/1000)
= 1.00
(ceiling plaster)= 20*(12.5/1000)
= 0.25
(flooring)= 0.22*(10/1000)
= 0.0022
Total Floor Finish load= 1.25 KN/m²

(iv)Live Load

i) RESIDENTIAL BUILDINS

a) Dwelling houses:

1) All rooms and kitchens	2'0	
2) Toilet and bath rooms	2·0	
3) Corridors, passages, staircases including tire escapes and store rooms	3·0	
4) Balconies	3.0	1·5 per r trated

b) Dwelling units planned and execut-cd in accordance with IS : 8888-1979* only:

1) Habitable rooms, kitchens, toilet and bathrooms	1·5	

3.2. DETERMINATION OF SEISMIC LOAD

SEISMIC COEFFICIENT AS PER NBC 105:2020	
Seismic Zoning Factor (Z) =	0.4
Importance Factor (I) =	1
Height of Building (h) =	12.5
Soil Type	D
Period of Vibration :	
For Reinforced Moment Resisting Frame Kt =	0.075
$T_1 = 1.25*k_t\, h^{0.75}$	0.623
Lower Period of the Flat Part of the Spectrum T_a =	0.5
Upper Period of the Flat Part of the Spectrum T_c =	2
Peak Spectral acceleration Normalized by PGA $_\alpha$ =	2.25
Coefficient to control the descending branch of the Spectrum K =	0.8
Calculation of Spectral Shape Factor : C_h (T)	
Since T_a<=T_1<=T_c	
Ch (T) =	2.25
Elastic Site Spectra for the Horizontal Loading (Clause 4.1.1 NBC 105:2020)	
C(T) = C_h (T) Z I =	0.900
Elastic Site Spectra for the SLS state (Clause 4.2 NBC 105:2020)	
$C_s(T_1)$ = 0.2 * C(T) =	0.180
Horizontal Base Shear Coefficient for Equivalent Static Method :	
Ref Table 5.2 NBC105:2020	
Ductility Factor for ULS state (R_u) =	4
Overstrength Factor for ULS state (Ω_u) =	1.5
Overstrength Factor for SLS state (Ω_s) =	1.25
Horizontal Base Shear Coefficient at the ULS state	
$C_d(T_1) = C(T)/(R_u*\Omega_u)$ =	0.150
Horizontal Base Shear Coefficient at the SLS state	
$C_d(T_1) = C_s(T_1)/(\Omega_s)$ =	0.144
Exponent related to the structural period (k) Clause 6.3 NBC105:2020=	1.061619

3.3 LOAD COMBINATIONS

Load combinations as per NBC 105:2020 are considered:

S.No.	For Concrete Design
1	1.2DL+1.5LL
2	DL+0.3LL±(EQX±0.3EQY)
3	DL+0.3LL±(0.3EQX±EQY)
4	DL+0.3LL±(RSX±0.3RSY)
5	DL+0.3LL±(0.3RSX±RSY) Note:

EQX/EQY = Equivalent Static Method
RSX/RSY = Response Spectrum Method

4. NUMERICAL MODELLING OF THE STRUCTURE

4.1. MODELLING APPROACH

The Structure is analyzed and designed using ETABS, a commercial software. It is widely used structural engineering software for analysis and design of building structures. It is equipped with the powerful analysis, design, graphics, and visualization capabilities. Attributes of the numerical model are:

- Frame elements of the structure have been modeled as linear elements with rigid end joints;
- Plinth level has been assumed as ground, where building has been assumed fixed;
- Slab and concrete wall elements are modeled as “thin shell” elements;
- Connection of slab and beam are provided through meshing of the elements;
- Staircase was modeled as part of the structure;
- Rigid diagramshave been assumed at each floor and roof level.

Figure 4provides snipping view of the numerical model.

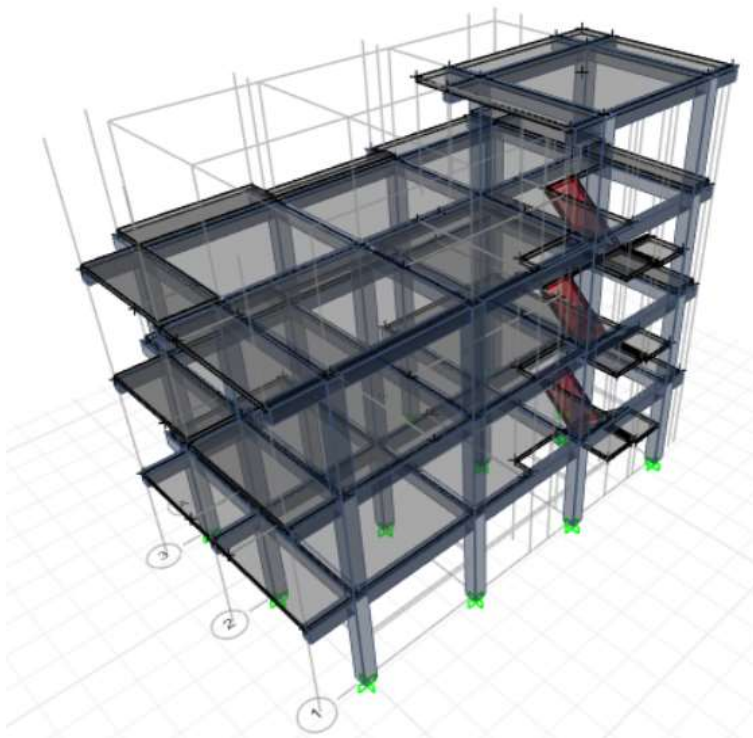


Figure 4 : 3D Model of the building in ETABS

4.2. PROPERTIES AND SECTIONS OF STRUCTURAL ELEMENTS:

Concrete has been assumed with isotropic properties with compressive strength 25 MPa (for all super structure). As recommended by the relevant Standards, cracked concrete section has been used as per 3.4 (NBC 105:2020). Refer Section 2.2.2 for section sizes.

S No.	Component	Flexural Stiffness	Shear Stiffness
1	Beam	0.35 Ec Ig	0.40 Ec Aw

2	Columns	0.70 Ec Ig	0.40 Ec Aw

4.3. INPUT LOADS IN ETABS

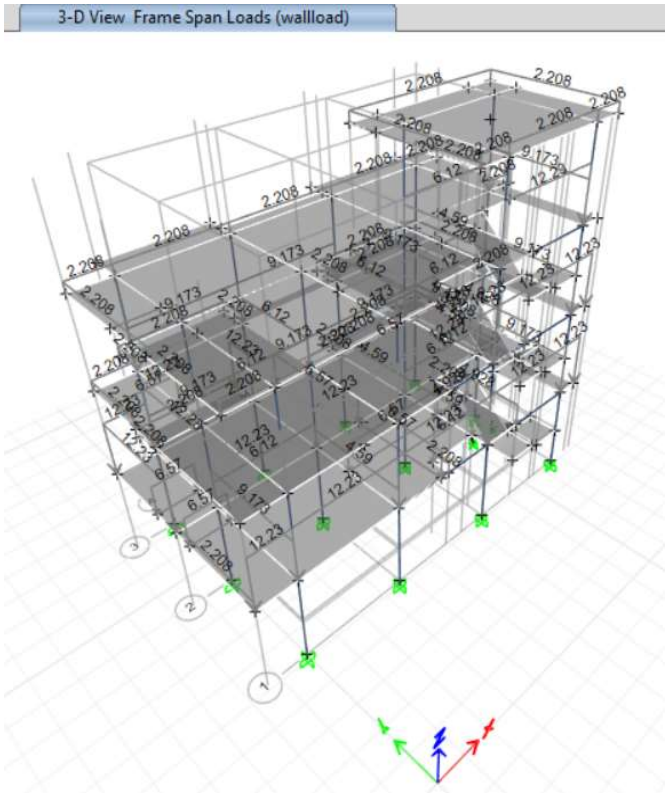


Figure 5 : Wall Load

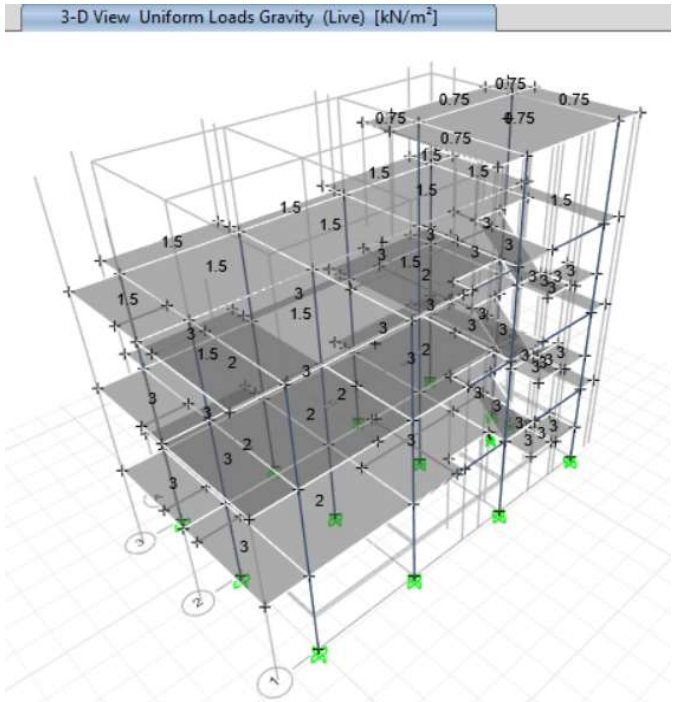


Figure 6 :LiveLoads

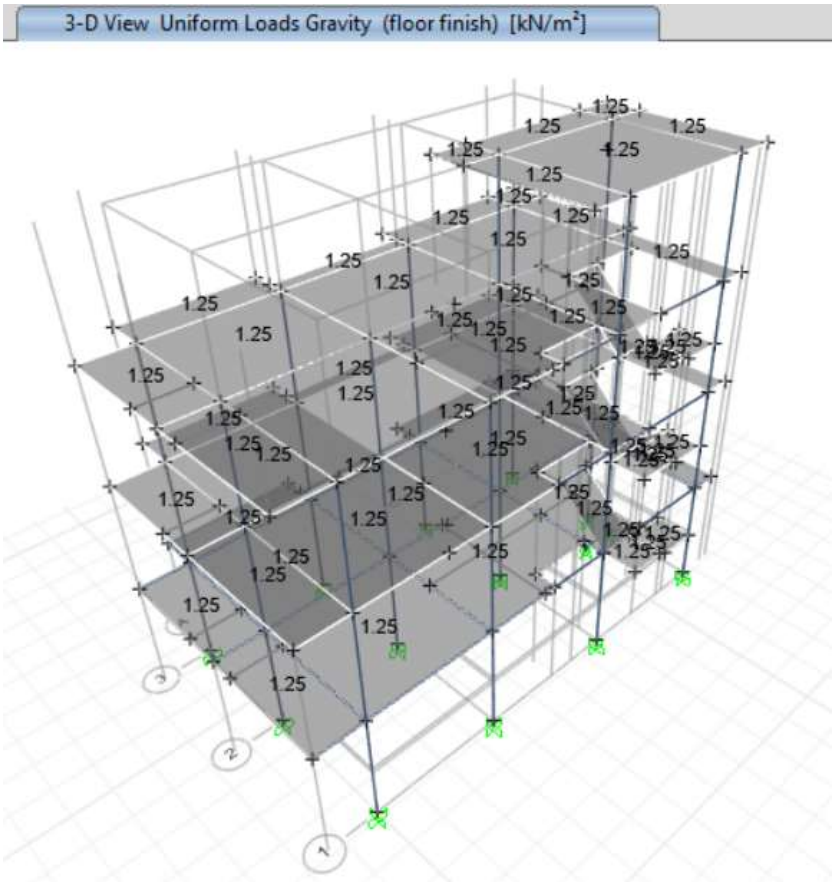


Figure 7 : Floor finishLoads

5. ANALYSIS OF BUILDING

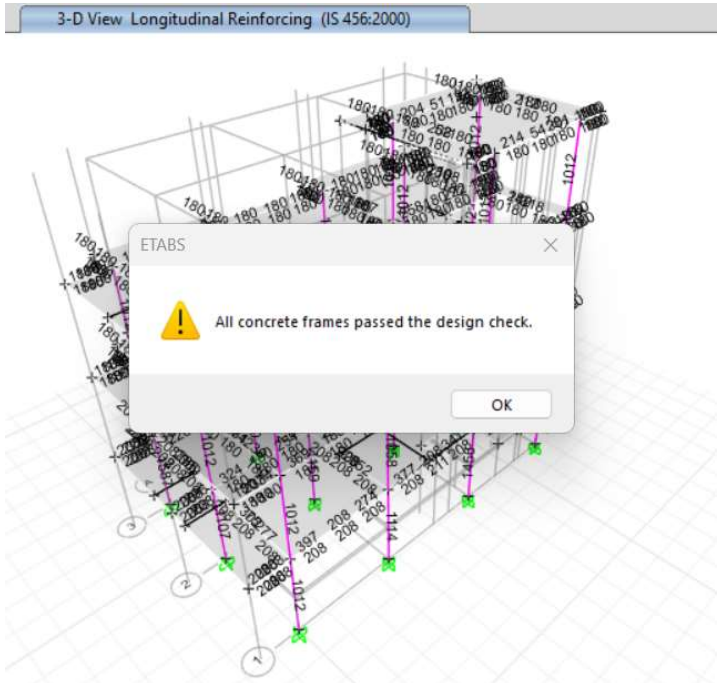


Figure 7 : Design Check

5.1. ANALYSIS RESULTS

TABLE: Base Reactions							
Output Case	Case Type	FX	FY	FZ	MX	MY	MZ
		kN	kN	kN	kN-m	kN-m	kN-m
eqx ULS	LinStatic	-633.0664	0	0	0.000000863	-4843.6673	5918.3353
eqy ULS	LinStatic	0	-633.0664	0	4843.6673	0	-4851.9513
eqx SLS	LinStatic	-607.7438	0	0	8.285E-07	-4649.9206	5681.6019
eqy SLS	LinStatic	0	-607.7438	0	4649.9206	0	-4657.8732

The table shows the comparable base reactions in seismic coefficient and response spectrum analysis with initial scale factor of 9.81.

TABLE: Load Pattern Definitions - Auto Seismic - User Coefficient							
Name	Ecc Ratio	Top Story	Bottom Story	C	K	Weight Used	Base Shear
						kN	kN
eqx SLS	0.1	4F	GF	0.144	1.061619	4220.443	607.7438
eqx ULS	0.1	4F	GF	0.15	1.061619	4220.443	633.0664
eqy SLS	0.1	4F	GF	0.144	1.061619	4220.443	607.7438
eqy ULS	0.1	4F	GF	0.15	1.061619	4220.443	633.0664

The table shows the value of C and K used according to NBC 105:2020.

TABLE: Modal Participating Mass Ratios											
Case	Mode	Period	UX	UY	UZ	SumUX	SumUY	SumUZ	RX	RY	RZ
		sec									
Modal	1	0.731	0.1266	0.6985	0	0.1266	0.6985	0	0.1493	0.0312	0.0062
Modal	2	0.686	0.5021	0.1166	0	0.6287	0.8151	0	0.0661	0.1107	0.2089
Modal	3	0.664	0.1882	0.0174	0	0.8169	0.8326	0	0.011	0.0738	0.6246
Modal	4	0.292	0.0772	0.0145	0	0.8941	0.8471	0	0.0793	0.3812	0.0101
Modal	5	0.287	0.016	0.0665	0	0.9101	0.9136	0	0.3655	0.0896	0.0207
Modal	6	0.251	0.000001195	0.0169	0	0.9101	0.9305	0	0.0892	0.000001099	0.0645
Modal	7	0.214	0.0157	0.0234	0	0.9258	0.9539	0	0.0893	0.0645	0.0181
Modal	8	0.188	0.0507	0.0156	0	0.9765	0.9694	0	0.064	0.1939	0.0001
Modal	9	0.171	0.0062	0.015	0	0.9827	0.9845	0	0.0523	0.0179	0.0313
Modal	10	0.14	0.0019	0.0106	0	0.9846	0.9951	0	0.0213	0.0039	0.0095
Modal	11	0.129	0.0154	0.0014	0	1	0.9965	0	0.003	0.0331	0.0003
Modal	12	0.119	0.00003606	0.0035	0	1	1	0	0.0096	0.0002	0.0057

Since the modal mass participation ratio is greater than 90% so it is satisfactory.

5.2. BUILDINGDRIFT

TABLE: Diaphragm Max Over Avg Drifts							
Story	Output Case	Case Type	Step Type	Item	Max Drift	Avg Drift	Ratio
4F	eqx ULS	LinStatic		Diaph D4 X	0.002808	0.002568	1.093
4F	eqy ULS	LinStatic		Diaph D4 Y	0.003128	0.002776	1.127
4F	eqx SLS	LinStatic		Diaph D4 X	0.002696	0.002466	1.093
4F	eqy SLS	LinStatic		Diaph D4 Y	0.003003	0.002665	1.127
3F	eqx ULS	LinStatic		Diaph D3 X	0.002798	0.002539	1.102
3F	eqy ULS	LinStatic		Diaph D3 Y	0.003496	0.002531	1.382
3F	eqx SLS	LinStatic		Diaph D3 X	0.002686	0.002438	1.102
3F	eqy SLS	LinStatic		Diaph D3 Y	0.003357	0.00243	1.382
2F	eqx ULS	LinStatic		Diaph D2 X	0.004007	0.003537	1.133
2F	eqy ULS	LinStatic		Diaph D2 Y	0.004521	0.003546	1.275
2F	eqx SLS	LinStatic		Diaph D2 X	0.003847	0.003396	1.133
2F	eqy SLS	LinStatic		Diaph D2 Y	0.004341	0.003404	1.275
1F	eqx ULS	LinStatic		Diaph D1 X	0.003034	0.002648	1.146
1F	eqy ULS	LinStatic		Diaph D1 Y	0.003749	0.002943	1.274
1F	eqx SLS	LinStatic		Diaph D1 X	0.002913	0.002542	1.146
1F	eqy SLS	LinStatic		Diaph D1 Y	0.003599	0.002825	1.274

Figure 7:Storey drift for Serviceability Limit State and Storey drift for Ultimate Limit State

The maximum storey drift for serviceability limit state was found to be 0.004241 which is within the limit i.e. 0.006 & the maximum storey drift for ultimate limit state was found to be 0.004521 which when multiplied by ductility factor Rμ (4 in this case) is 0.018084and is within the limit i.e. 0.025.

TABLE: Story Max Over Avg Displacements							
Story	Output Case	Case Type	Step Type	Direction	Maximum	Average	Ratio
					mm	mm	
4F	eqx ULS	LinStatic		X	35.907	33.483	1.072
3F	eqx ULS	LinStatic		X	30.15	26.967	1.118
2F	eqx ULS	LinStatic		X	21.619	19.137	1.13
1F	eqx ULS	LinStatic		X	9.482	8.276	1.146
4F	eqy ULS	LinStatic		Y	45.616	41.965	1.087
3F	eqy ULS	LinStatic		Y	35.841	28.315	1.266
2F	eqy ULS	LinStatic		Y	25.377	20.342	1.248
1F	eqy ULS	LinStatic		Y	11.714	9.196	1.274
4F	eqx SLS	LinStatic		X	34.471	32.144	1.072
3F	eqx SLS	LinStatic		X	28.944	25.888	1.118
2F	eqx SLS	LinStatic		X	20.754	18.371	1.13
1F	eqx SLS	LinStatic		X	9.103	7.945	1.146
4F	eqy SLS	LinStatic		Y	43.792	40.286	1.087
3F	eqy SLS	LinStatic		Y	34.408	27.182	1.266
2F	eqy SLS	LinStatic		Y	24.362	19.528	1.248
1F	eqy SLS	LinStatic		Y	11.246	8.828	1.274

Figure9 :Storeydisplacement for Ultimate Limit State and Serviceability Limit State

5.3. Support Reaction for Foundation Design (Combination: DL+LL)

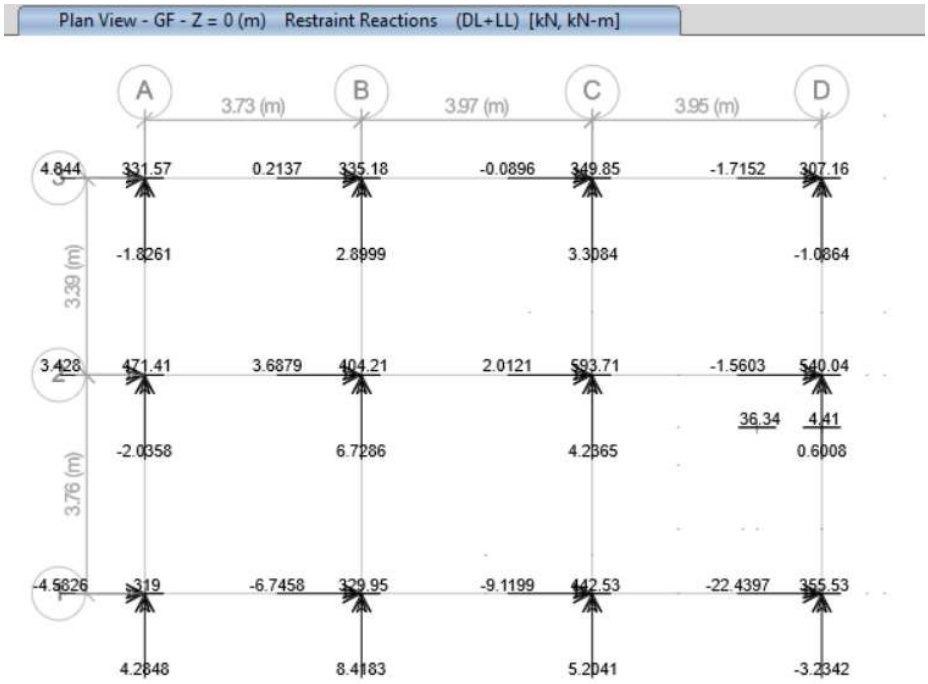


Figure 9 : Support reactions for foundation design

5.4. MEMBER FORCES

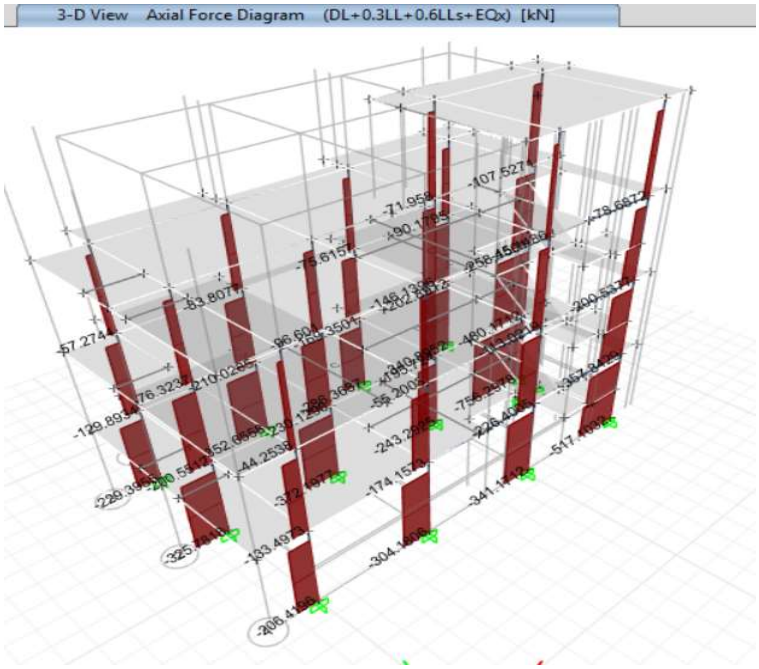


Figure 8 : Sample axial force diagram

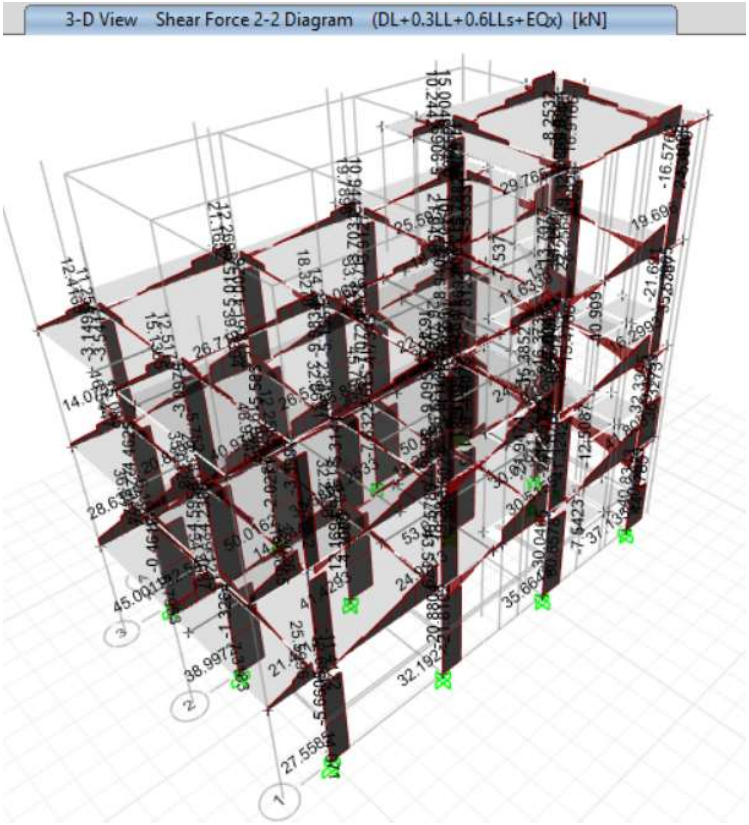


Figure 9 : Sample shear force diagram

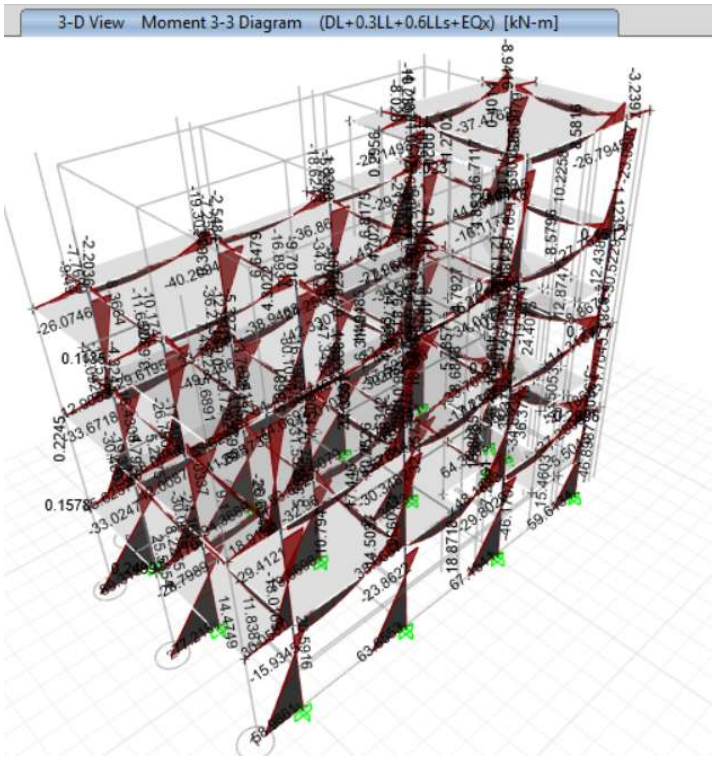
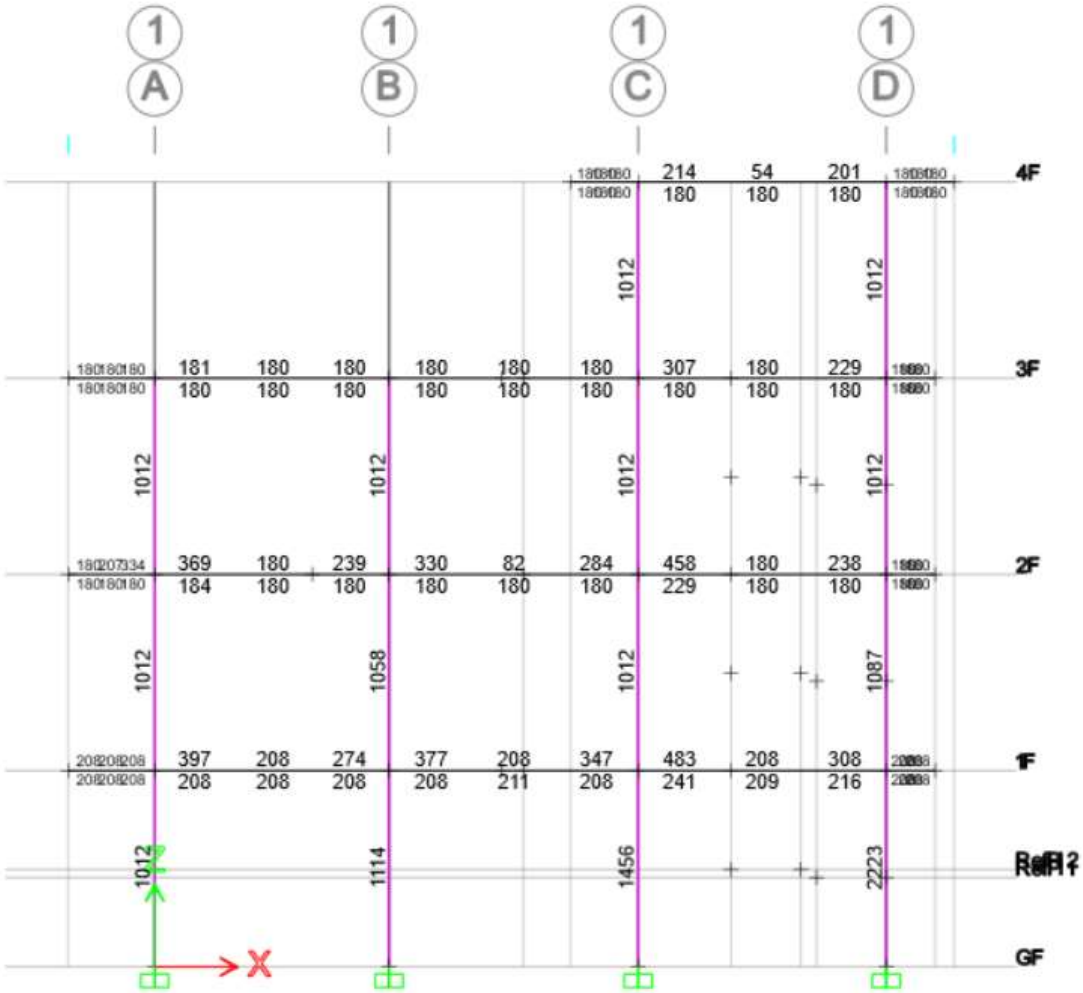


Figure 10 : Sample bending moment diagram

5.5. DESIGN OUTPUT

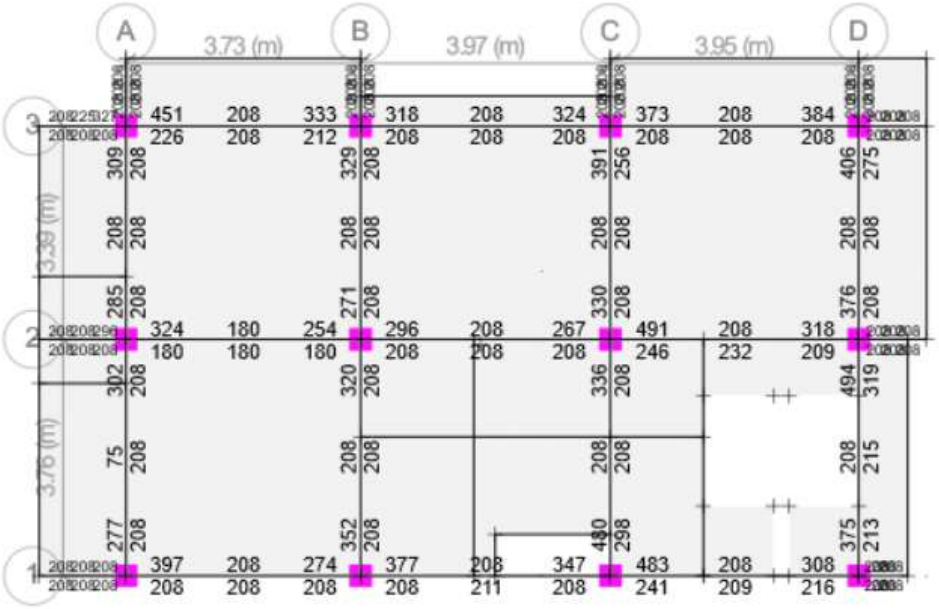
Elevation View - 1 Longitudinal Reinforcing (IS 456:2000)



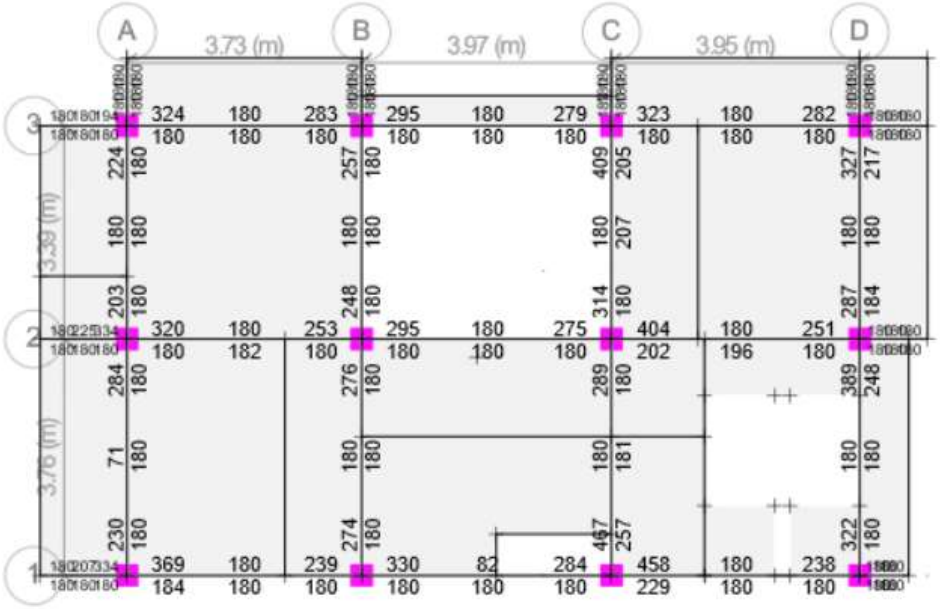
Elevation View - 2 Longitudinal Reinforcing (IS 456:2000)

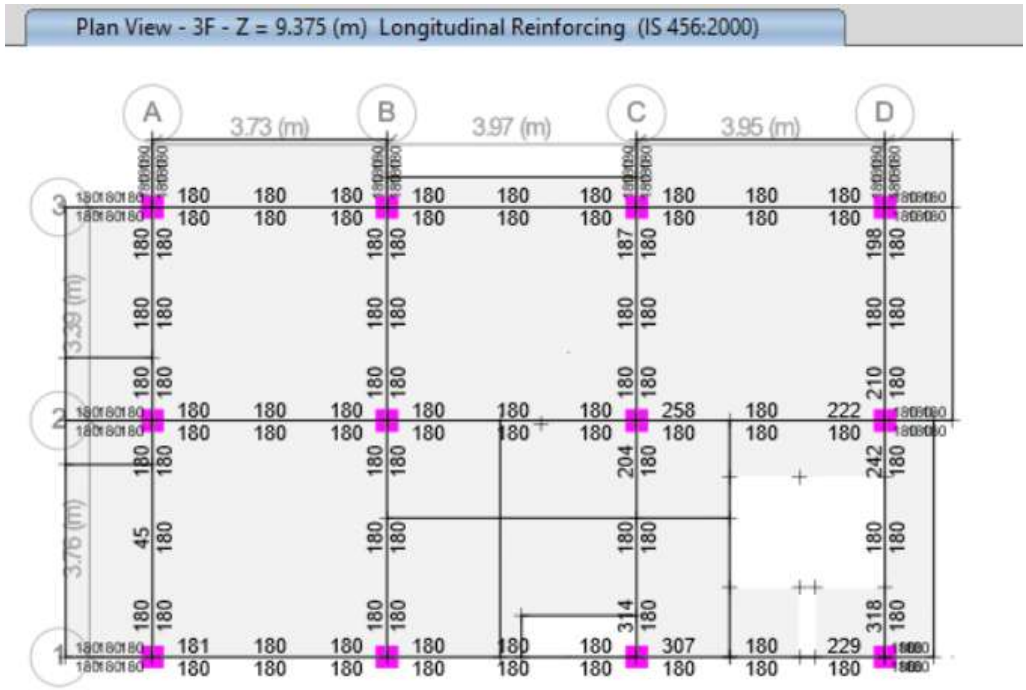
Elevation View - 3 Longitudinal Reinforcing (IS 456:2000)

Plan View - 1F - Z = 3.125 (m) Longitudinal Reinforcing (IS 456:2000)



Plan View - 2F - Z = 6.25 (m) Longitudinal Reinforcing (IS 456:2000)





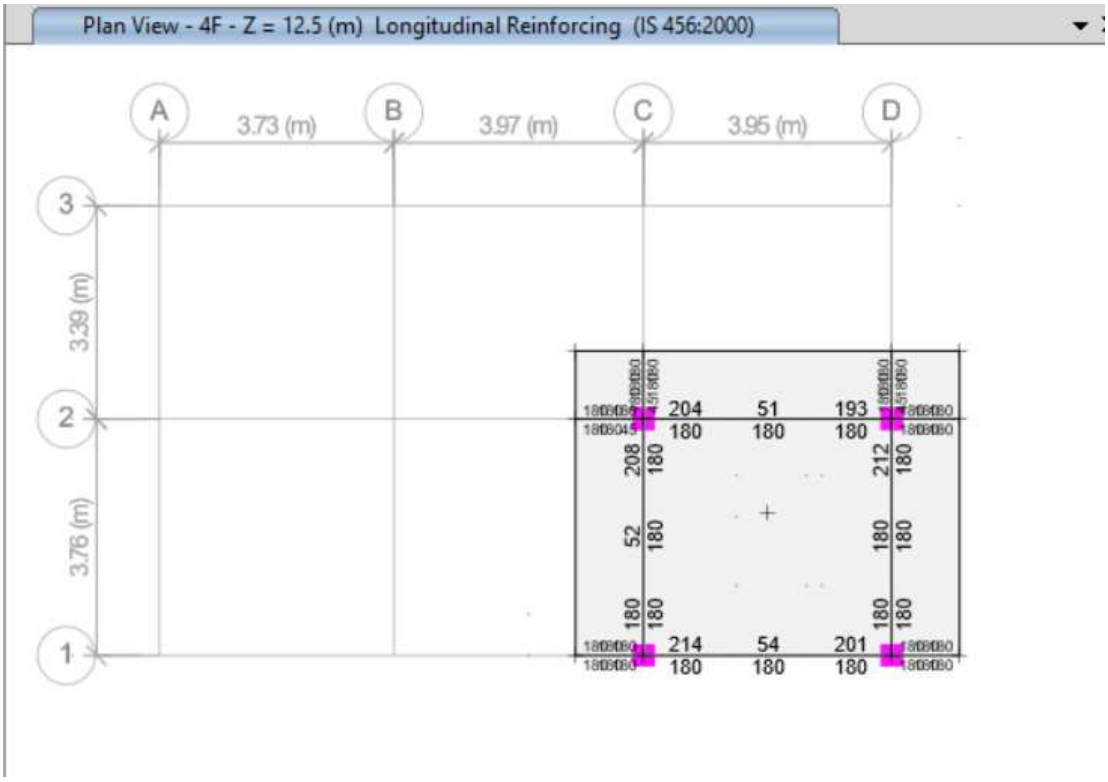


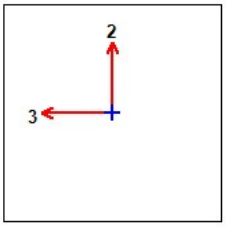
Figure 11 : Rebar Result Details

6. DESIGN OF STRUCTURAL ELEMENTS

6.1. DESIGN OF COLUMNS

ETABS Concrete Frame Design

IS 456:2000 + IS 13920:2016 Column Section Design (Summary)



Column Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
1F	C1	36	column 14*14	DL+0.3LL+0.6LLs-EQx	0	3125	0.73	Ductile Frame

Section Properties

b (mm)	h (mm)	dc (mm)	Cover (Torsion) (mm)
355.6	355.6	56	30

Material Properties

E_c (MPa)	f_{ck} (MPa)	Lt.Wt Factor (Unitless)	f_y (MPa)	f_{ys} (MPa)
25000	25	1	500	500

Design Code Parameters

γ_c	γ_s
1.5	1.15

Axial Force and Biaxial Moment Design For P_u , M_{u2} , M_{u3}

Design P_u kN	Design M_{u2} kN-m	Design M_{u3} kN-m	Minimum M_2 kN-m	Minimum M_3 kN-m	Rebar Area mm²	Rebar % %
459.0404	-16.5256	-87.3558	9.1808	9.1808	1084	0.86

Axial Force and Biaxial Moment Factors

	K Factor Unitless	Length mm	Initial Moment kN-m	Additional Moment kN-m	Minimum Moment kN-m
Major Bend(M3)	0.661938	2718.6	-36.3636	0	9.1808
Minor Bend(M2)	0.629094	2718.6	-6.9635	0	9.1808

Shear Design for V_{u2} , V_{u3}

	Shear V_u kN	Shear V_c kN	Shear V_s kN	Shear V_p kN	Rebar A_{sv} /s mm²/m
Major, V_{u2}	46.892	75.3667	42.6146	26.8083	394.16

	Shear V_u kN	Shear V_c kN	Shear V_s kN	Shear V_p kN	Rebar A_{sv} /s mm²/m
Minor, V_{u3}	46.1652	75.3667	42.6146	46.1652	394.16

Joint Shear Check/Design

	Joint Shear Force kN	Shear V_{Top} kN	Shear $V_{u,Tot}$ kN	Shear V_c kN	Joint Area cm²	Shear Ratio Unitless
Major Shear, V_{u2}	N/N	N/N	N/N	N/N	N/N	N/N
Minor Shear, V_{u3}	N/N	N/N	N/N	N/N	N/N	N/N

(1.4) Beam/Column Capacity Ratio

Major Ratio	Minor Ratio
N/N	N/N

Additional Moment Reduction Factor k (IS 39.7.1.1)

A_g cm²	A_{sc} cm²	P_{uz} kN	P_b kN	P_u kN	k Unitless
1264.5	10.8	1828.9983	568.607	459.0404	1

Additional Moment (IS 39.7.1)

	Consider M_a	Length Factor	Section Depth (mm)	KL/Depth Ratio	KL/Depth Limit	KL/Depth Exceeded	M_a Moment (kN-m)
Major Bending (M_3)	Yes	0.87	355.6	5.061	12	No	0
Minor Bending (M_2)	Yes	0.87	355.6	4.809	12	No	0

Notes:

N/A: Not Applicable

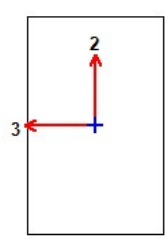
N/C: Not Calculated

N/N: Not Needed

6.2. DESIGN OF BEAM

ETABS Concrete Frame Design

IS 456:2000 + IS 13920:2016 Beam Section Design (Summary)



Beam Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
1F	B18	104	B-10*16	DL+0.3LL+0.6LLs-EQx	177.8	3950	1	Ductile Frame

Section Properties

b (mm)	h (mm)	b_f (mm)	d_s (mm)	d_{ct} (mm)	d_{cb} (mm)
254	406.4	254	0	25	25

Material Properties

E_c (MPa)	f_{ck} (MPa)	Lt.Wt Factor (Unitless)	f_y (MPa)	f_{ys} (MPa)
22360.68	20	1	500	500

Design Code Parameters

γ_c	γ_s
1.5	1.15

Factored Forces and Moments

Factored M_{u3} kN-m	Factored T_u kN-m	Factored V_{u2} kN	Factored P_u kN
-68.4677	1.6356	88.6649	0

Design Moments, M_{u3}& M_t

Factored Moment kN-m	Factored M_t kN-m	Positive Moment kN-m	Negative Moment kN-m
-68.4677	2.5016	0	-70.9693

Design Moment and Flexural Reinforcement for Moment, M_{u3}&T_u

	Design -Moment kN-m	Design +Moment kN-m	-Moment Rebar mm²	+Moment Rebar mm²	Minimum Rebar mm²	Required Rebar mm²
Top (+2 Axis)	-70.9693		491	0	491	208
Bottom (-2 Axis)		0	246	0	0	246

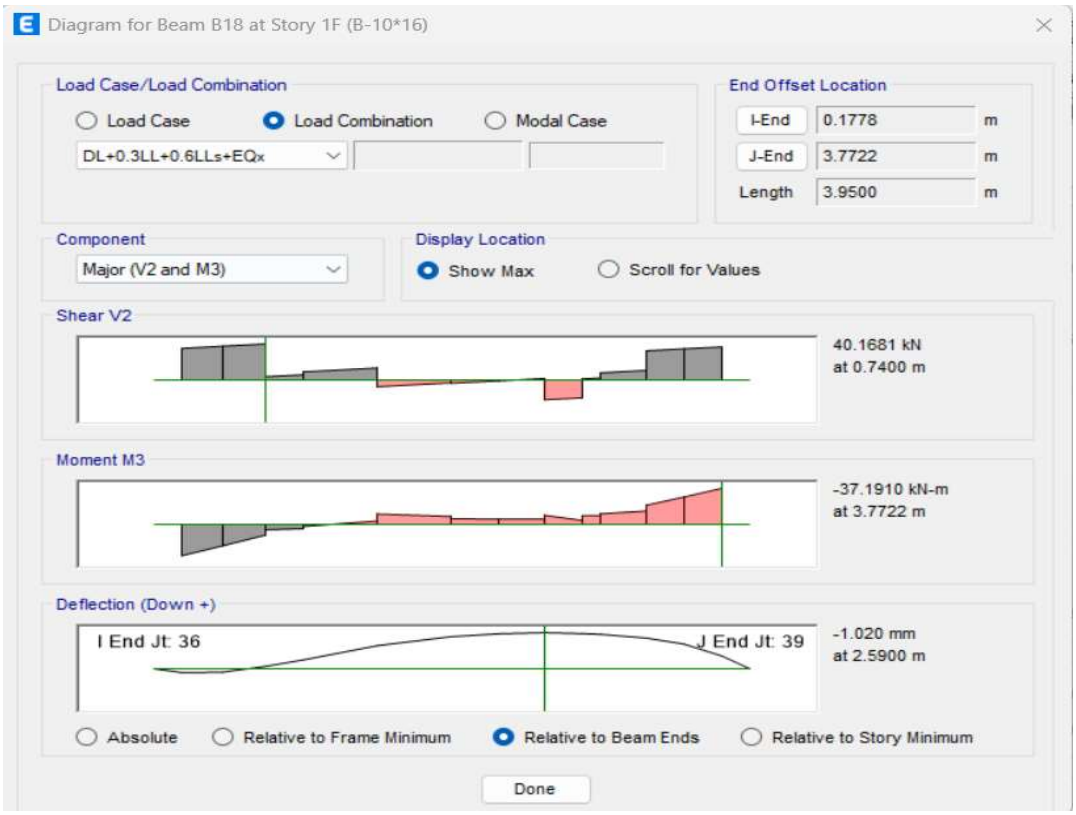
Shear Force and Reinforcement for Shear, V_{u2}&T_u

Shear V_e kN	Shear V_c kN	Shear V_s kN	Shear V_p kN	Rebar A_{sv} /s mm²/m
88.6649	0	98.9681	41.9457	719.06

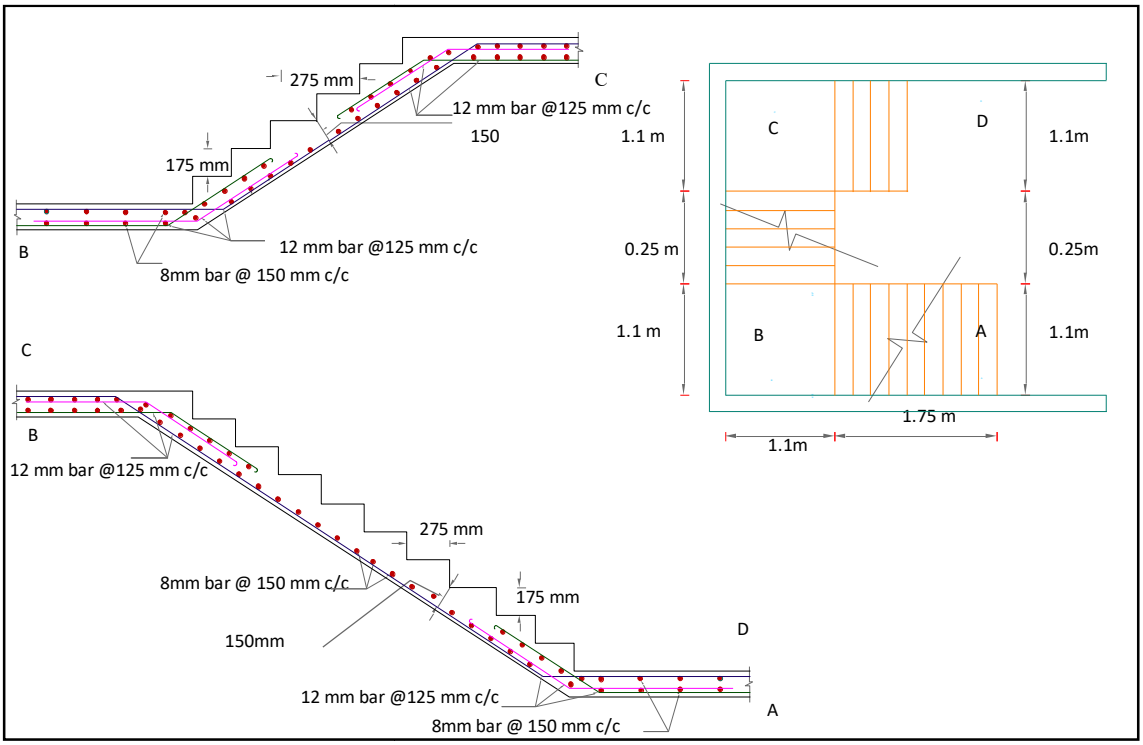
Torsion Force and Torsion Reinforcement for Torsion, T_u& V_{U2}

T_u kN-m	V_u kN	Core b_1 mm	Core d_1 mm	Rebar A_{svt} /s mm²/m
1.6356	88.6649	224	376.4	379.64

Deflection of Longest Beam



6.3. DESIGN OF STAIRCASE



6.4. DESIGN OF SLAB

SAMPLE DESIGN OF FLOOR SLAB

Design Data

Dimensions of the slab (c/c distance b/w supports), f_{ck} = **20** N/mm^2

Length of short span, L_x = **3.76** m f_y = **500** N/mm^2

Length of long span, L_y = **3.97** m

Width of the supporting beam, = **230** mm

Clear cover to main reinforcement = **20** mm

Assume dia. of reinforcement steel = **8** mm

Calculations

Assume the thickness of slab as **125** mm ; Effective depth, d = 101 mm

Effective span, l_x = 3.76 m (or) 3.631 m whichever is less; lx = 3.631 m

l_y = 3.97 m (or) 3.841 m whichever is less; ly = 3.841 m

(l_y / l_x) = 1.06 < 2 ; Here, (l_y / l_x) is less than 2, Hence design the slab as two way slab

Load Calculations

Dead Load of slab = 0.125 x 25 = 3.13 KN/m^2

Finishes load on slab = **1.50** KN/m^2

Live Load on slab = **3.0** KN/m^2

Total Dead load acting on the Structure = 4.63 KN/m^2

Total live load acting on the Structure = 3.0 KN/m^2

Factored Design Load w = 11.44 KN/m^2

Support Condition (Type of panel according to support condition)

Two Long Edges Discontinuous

Short span coefficient for (l_y / l_x) = 1.06,

For negative moment, a_x = 0.0000

For positive moment, a_x = 0.0398

For this support condition,

Long span coefficient,

For negative moment, a_y = 0.045

For positive moment, a_y = 0.035

Moment Calculation

Max. BM per unit width, $M_x = a_x\, w l_x^2$ & $M_y = a_y\, w l_x^2$

	M_u KNm	M_u / bd^2 N/mm^2	p_t %	$A_{st, req}$ mm^2
For Short Span,				
At mid span,	6.00	-	0.14	141.533
At supports,	0.00	-	0	0
For Long span,				
At mid span,	5.28	-	0.12	123.991
At supports,	6.79	-	0.16	160.97

$A_{st, min}$ = (0.12/100) bD = 150 mm^2

Reinforcement details

Provide Y **8** @ **150** mm c/c at midspan & supports for short span (A_{st} pro. = 335 mm^2)

Provide Y **8** @ **150** mm c/c at midspan & supports for long span (A_{st} pro. = 335 mm^2)

Check for Deflection

Percentage of tension reinforcement = 0.33 %

f_s = 0.58 f_y ($A_{st\ req}$ / $A_{st\ pro}$) = 122.47684

Refer Fig. 4 of IS 456,

Modification factor = **1.7**

Allowable (Span / d_{eff}) ratio = **39.1**

Effective depth required = 93 mm

< d prov.

Hence OK

Summary:

Overall Depth of slab = 125 mm

Main Bar:

Astx +ve = 8 Ø @ 150 mm c/c

Astx -ve = 8 Ø @ 150 mm c/c

Asty +ve = 8 Ø @ 150 mm c/c

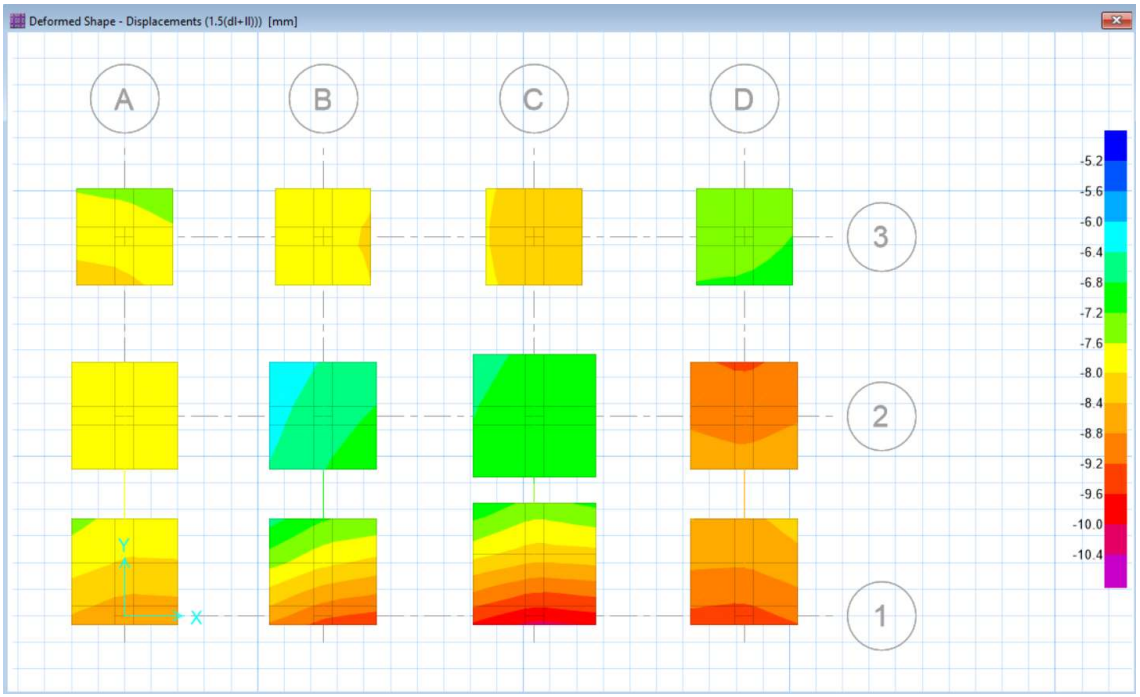
Asty -ve = 8 Ø @ 150 mm c/c

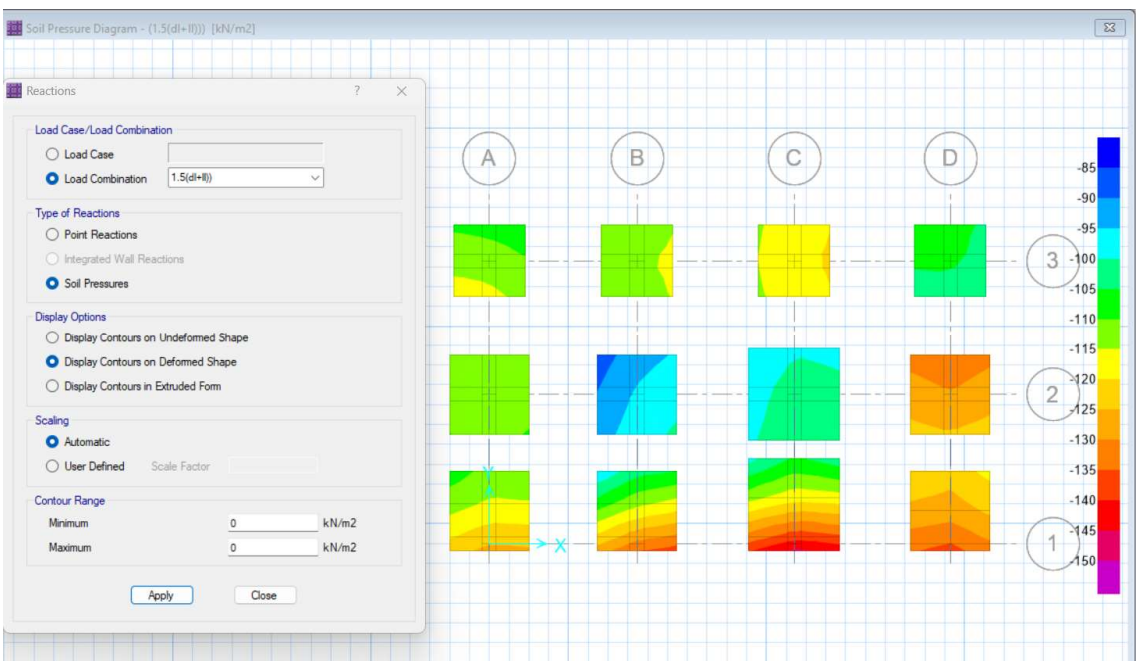
6.5. DESIGN OF SHEAR WALL

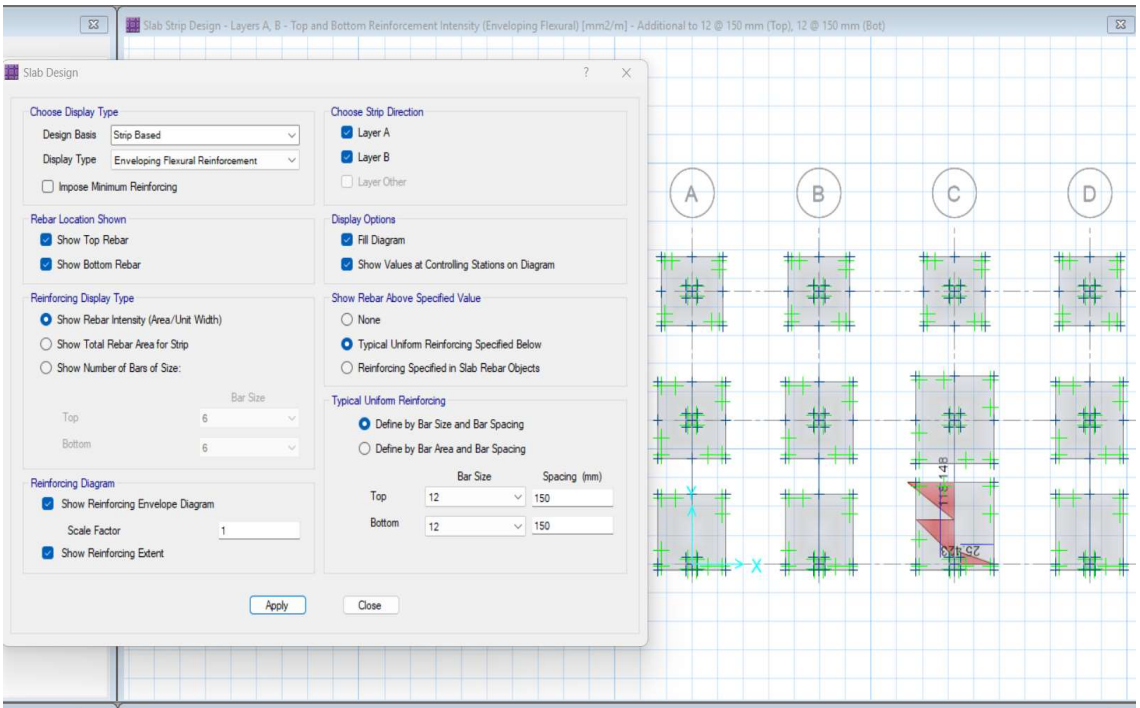
No Shear wall in this Building

6.6. FOUNDATION DESIGN

The analysis and design of the building have been carried out using a finite element software Safe 2016.



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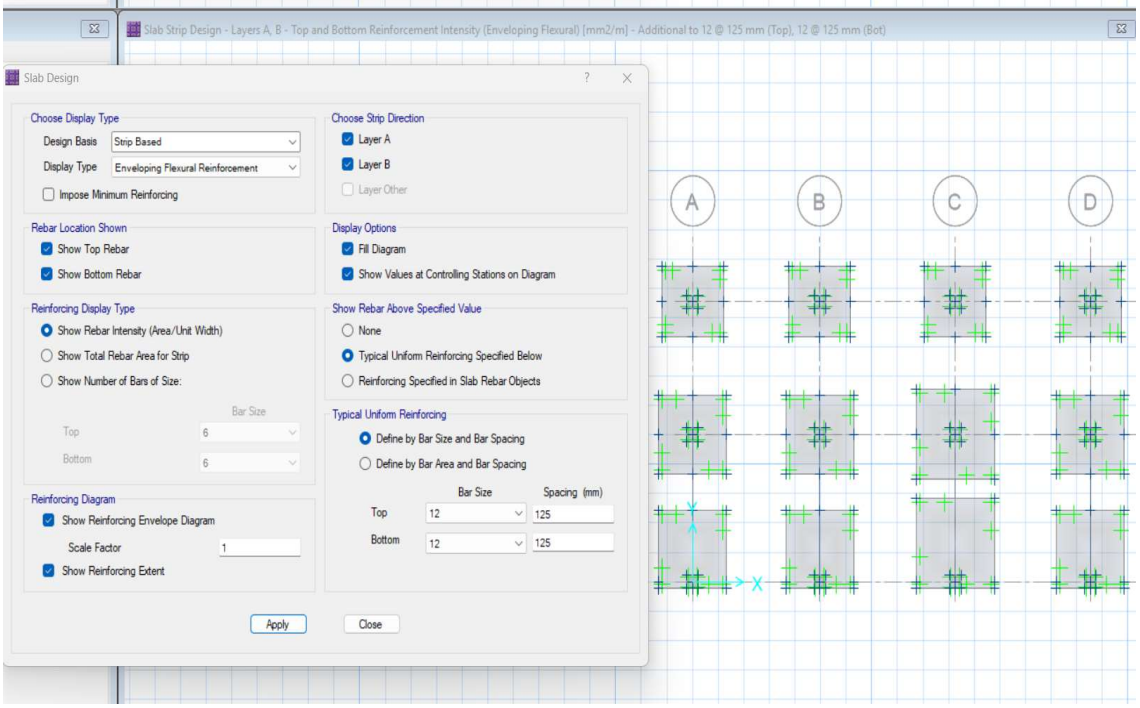
Choose Display Type
Design Basis
Strip Based
Display Type
Enveloping Flexural Reinforcement
Impose Minimum Reinforcing
Choose Strip Direction
Layer A
Layer B
Layer Other
Rebar Location Shown
Show Top Rebar
Show Bottom Rebar
Display Options
Fill Diagram
Show Values at Controlling Stations on Diagram
Reinforcing Display Type
Show Rebar Intensity (Area/Unit Width)
Show Total Rebar Area for Strip
Show Number of Bars of Size:
Bar Size
Top
6
Bottom
6
Show Rebar Above Specified Value
None
Typical Uniform Reinforcing Specified Below
Reinforcing Specified in Slab Rebar Objects
Typical Uniform Reinforcing
Define by Bar Size and Bar Spacing
Define by Bar Area and Bar Spacing
Bar Size
Spacing (mm)
Top
12
150
Bottom
12
150
Reinforcing Diagram
Show Reinforcing Envelope Diagram
Scale Factor
1
Show Reinforcing Extent
Apply
Close
A
B
C
D
X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Choose Display Type
Design Basis
Strip Based
Display Type
Enveloping Flexural Reinforcement
Impose Minimum Reinforcing
Choose Strip Direction
Layer A
Layer B
Layer Other
Rebar Location Shown
Show Top Rebar
Show Bottom Rebar
Display Options
Fill Diagram
Show Values at Controlling Stations on Diagram
Reinforcing Display Type
Show Rebar Intensity (Area/Unit Width)
Show Total Rebar Area for Strip
Show Number of Bars of Size:
Bar Size
Top
6
Bottom
6
Show Rebar Above Specified Value
None
Typical Uniform Reinforcing Specified Below
Reinforcing Specified in Slab Rebar Objects
Typical Uniform Reinforcing
Define by Bar Size and Bar Spacing
Define by Bar Area and Bar Spacing
Bar Size
Spacing (mm)
Top
12
125
Bottom
12
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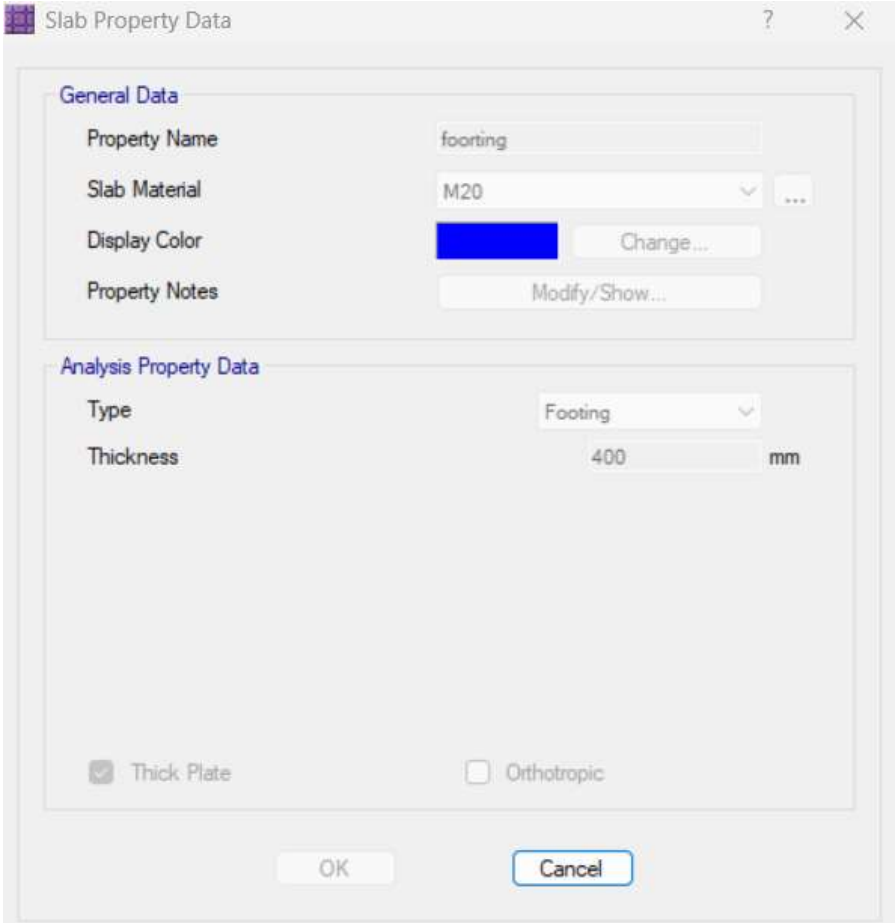
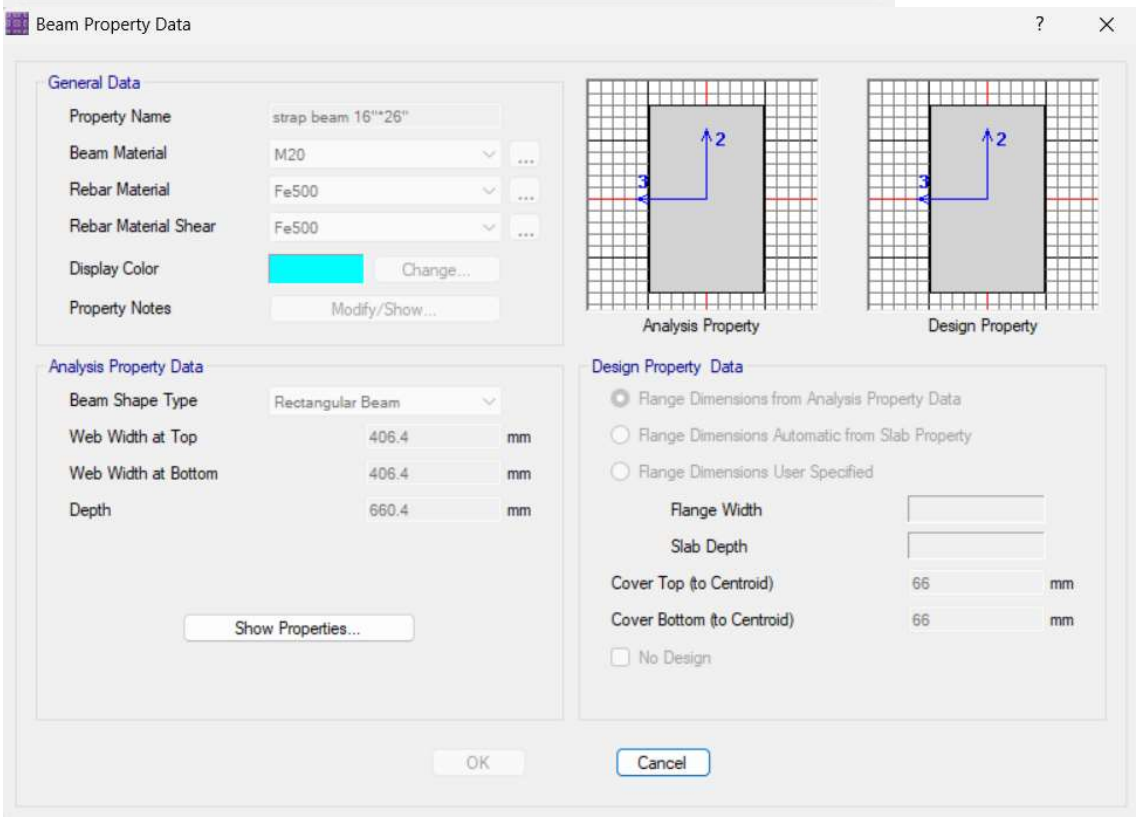
Reinforcing Diagram
Show Reinforcing Envelope Diagram
Scale Factor
1
Show Reinforcing Extent
Apply
Close
A
B
C
D
X


Slab Property Data
General Data
Property Name
foorting
Slab Material
M20
Display Color
Change...
Property Notes
Modify/Show...
Analysis Property Data
Type
Footing
Thickness
400
mm
Thick Plate
Orthotropic
OK
Cancel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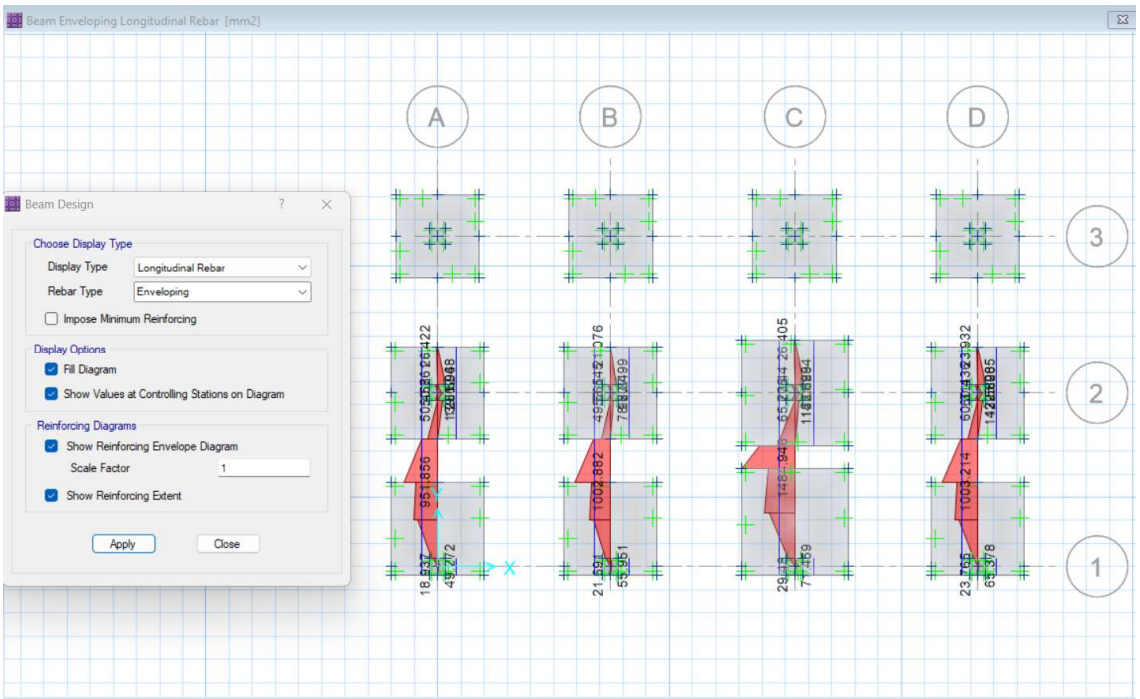
General Data
Property Name
strap beam 16"*26"
Beam Material
M20
Rebar Material
Fe500
Rebar Material Shear
Fe500
Display Color
Change...
Property Notes
Modify/Show...
Analysis Property
Design Property
Analysis Property Data
Beam Shape Type
Rectangular Beam
Web Width at Top
406.4
mm
Web Width at Bottom
406.4
mm
Depth
660.4
mm
Show Properties...
Design Property Data
Flange Dimensions from Analysis Property Data
Flange Dimensions Automatic from Slab Property
Flange Dimensions User Specified
Flange Width
Slab Depth
Cover Top (to Centroid)
66
mm
Cover Bottom (to Centroid)
66
mm
No Design
OK
Cancel



Summary For A1-A2, B1-B2 & D1-D2					
Footing Slab	Length (m	Width(m)	Thickness		
Edge	2	2	0.40	Flat	Depth=2.0
Central	2	2	0.40		
Reniforcement	12mm dia rebar @150mm c/c spacing BOTHWAY				
Strap Beam					
width	0.406				
depth	0.66				
Longitudinal reinforcement					
At top Provide 5 nos. of 16 mm bar as longitudinal reinforcement					
At Bottom Provide 5 nos. of 12mm bar as longitudinal reinforcement					
Shear reinforcement 4 legged 8 mm dia rebar @150mm c/c spacing					
Summary For C1-C2					
Footing Slab	Length (m	Width	Thickness		
Edge	2.3	2.3	0.4	Flat	Depth=2.0
Central	2.3	2.3	0.4		
Reniforcement	12mm dia rebar @125mm c/c spacing BOTHWAY				
Strap Beam					
width	0.406				
depth	0.66				
Longitudinal reinforcement					
At top Provide 5 nos. of 20 mm bar as longitudinal reinforcement					
At Bottom Provide 5 nos.of 16 mm as longitudinal reinforcement					
Shear reinforcement 4 legged 8mm dia rebar @150mm c/c spacing					
Summary of isolated footing A3, B3, C3 & D3					
Footing Slab	Length (m	Width(m)	Thickness		
Edge	1.8	1.8	0.40	Flat	Depth=2.0
Central	1.8	1.8	0.40		
Reniforcement	12mm dia rebar @150mm c/c spacing BOTHWAY				

7. SUMMARY OF FINAL DESIGN OUTPUTS

Section of Floor Columns& Reinforcement details

SN	Node	column size	Ground Floor	First Floor	Second Floor	Stair cover	Stirrupus
			Area of rebar provided	Area of rebar provided	Area of rebar provided	Area of rebar provided	
1	A1	14"*14"	8-16mm	4-12mm+4-16mm	4-12mm+4-16mm		As per norms
2	A2		8-16mm	4-12mm+4-16mm	4-12mm+4-16mm		
3	A3		8-16mm	4-12mm+4-16mm	4-12mm+4-16mm		
4	B1		8-16mm	4-12mm+4-16mm	4-12mm+4-16mm		
5	B2		8-16mm	4-12mm+4-16mm	4-12mm+4-16mm		
6	B3		4-20mm+4-16mm	8-16mm	4-12mm+4-16mm		
7	C1		8-16mm	8-16mm	4-12mm+4-16mm	4-12mm+4-16mm	
8	C2		4-20mm+4-16mm	4-20mm+4-16mm	8-16mm	4-12mm+4-16mm	
9	C3		4-20mm+4-16mm	8-16mm	4-12mm+4-16mm		
10	D1		8-20mm	4-20mm+4-16mm	8-16mm	4-12mm+4-16mm	
11	D2		4-20mm+4-16mm	8-16mm	4-12mm+4-16mm	4-12mm+4-16mm	
12	D3		4-20mm+4-16mm	8-16mm	4-12mm+4-16mm		

Section of Floor Beams & Reinforcement details

SN	Designation	size	Top bar	Bottom bar	Stirrupus
			Area of rebar provided	Area of rebar provided	
1	Beam				As per norms
Along All grid	first floor beam level and cantilever part	230mm*400mm	3-16mm (T)	3-16mm (T)	
Along All grid	Second floor beam level and cantilever part	230mm*355mm	3-16mm (T)	3-16mm (T)	
Along All grid	Third floor beam level and cantilever part	230mm*355mm	3-16mm (T)	3-16mm (T)	
Along All grid	Top floor beam level and cantilever part	230mm*355mm	3-12mm (T)	3-12mm (T)	
2	Foundation tie beam	230mm*230mm	2-16mm	2-16mm	
3	Plinth beam	230mm*230mm	2-16mm	2-16mm	

Annex 1: Relevant Code followed

The structural calculation of this building shall be in accordance with all applicable laws and regulations of the Government of Nepal, and its referral to any other standards. A summary of the codes and standards in used in structural analysis and design have been listed as below.

S. No.	Codes and Standards	Description
Dead and Live Load Determination		
	IS 875 (Part 1): 1987	Code of Practice for Design Loads (other than Earthquake) for Buildings and Structures: Part 1 Dead Loads – Unit Weights of Building Material and Stored Materials (Second Revision)
	IS 875 (Part 2): 1987	Code of Practice for Design Loads (other than Earthquake) for Buildings and Structures: Part 2 Imposed Loads (Second Revision)
	IS 875 (Part 3): 1987	Code of Practice for Design Loads (other than Earthquake) for Buildings and Structures: Part 3 Wind Loads (Second Revision)
Earthquake Load Calculation		
	NBC 105:2020	Seismic Design of Buildings in Nepal
Concrete Design and Detailing		
	IS 456: 2000	Plain and Reinforced Concrete - Code of Practice
	SP 34: 1987	Handbook on Concrete Reinforcement and Detailing
	IS 13920: 2016	Ductile detailing of reinforced concrete structures subjected to seismic forces - Code of practice
	NBC201:1994	Mandatory Rules of Thumbs - Reinforced Concrete Building with Masonry Infill.
	NBC110-1994	Plain and Reinforced Concrete.
Specification of Materials		
	IS 383: 1970	Specification for coarse and fine aggregates from natural sources for concrete
	IS 1786: 1985	Specification for high strength deformed steel bars and wires for concrete reinforcement (superseding IS:1139 - 1966)
	NS191-2046BS	Deformed steel bars and wires for concrete reinforcement
	IS801:1975	Code of practice of use of cold-formed light gauge steel structural members in general building construction.

Remarks:

Hence, the analysis and design were completed by rigorous calculations using the ETABS software. Using the knowledge of recent design codes, byelaws and other practical aspects like experience, judgment, etc. in conjunction with conceptual design; the structure is designed for safety and economy. The designer will not be responsible if any alterations to the structural system is made by Supervisor or the contractor without the prior written permission from the designer, or the alterations to the non-structural system is made such that the weight. The structure was safe up to the designed story.

Note:The site supervisions & quality test of the different constructions materials of the proposed building do not undertaken by the Designer. Hence any failure or error occurred with the building due to the non – implementation of the recommendations made in this report and structural drawings or due to improper construction will not be the responsibility of the Designer. Also, the structural analysis has been completed on the assumptions that the proposed building shall be constructed under the proper supervisions of skilled technical person.